LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 4

1. What is the relationship between symbol rate and bit rate? $N = S \cdot r$

2. Study the works of Shannon and Nyquist on channel capacity. Each places an upper limit on the bit rate of a channel based on two different approaches. How are the two related?

   $C = B \cdot \log_2(1+SNR)$

   $N = 2B \cdot \log_2 L$

3. How are binary values represented in amplitude shift keying, and what is the limitation of this approach? binary values are represented by different amplitude of carrier frequency. Affected by noise.

4. What is QAM?

   The Amplitude and phase of carrier signals are both be uttered to modulate the message signal.

5. What does the sampling theorem tell us concerning the rate of sampling required for an analogue signal?

   sampling rate $\geq 2\times$ highest frequency

6. Consider a noiseless channel with a bandwidth of 125 KHz. Assume that QPSK is used. What is the theoretical maximum bit rate that we can send through this channel? What is the theoretical maximum symbol rate? $r=2$

   $2\times125\times10^3\times2 = 500000$ bps

   250000 baud

7. Consider a noisy channel with a bandwidth of 125 KHz. Assume that the average received power is 7 watts and the average AWGN noise power is 1 watt. What is the theoretical maximum bit rate for this channel? $SNR = 7$

   $C = 125\times10^3\times\log_2(1+7) = 3.75\times10^5$ bps

8. An analog signal has a bit rate of 8000 bps and a symbol rate of 1000 symbols per second. How many bits are carried by each signal element (symbol)? How many different symbols do we need? $8000/1000 = 8$ $r=8$

   $L = 2^8 = 256$

9. Suppose for a binary digital modulation, e.g., BPSK, the symbol rate is 600 Kbaud, then what is the bit rate? $r=1$ $N = 600$ bps

10. If the baud rate (symbol rate) is 600 Kbaud for an 8PSK signal, then what is the bit rate?

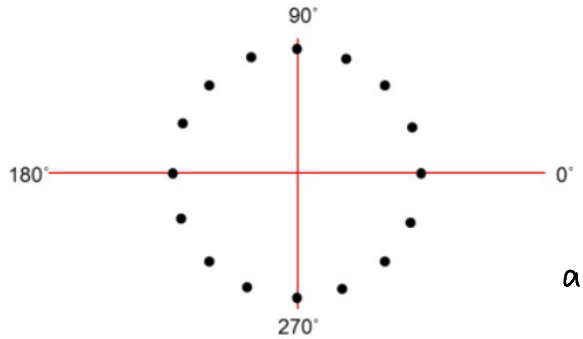
    L的值 $r=3$ 1800 Kbps

11. Using the signal constellation diagram shown below, answer the following questions.
    a) What type of modulation does this represent?
    b) How many symbols are represented? 16PSK
    c) How many bits per symbol are used? 16
    d) If the Baud rate is 10,000 symbols/second, what is the bit rate?

    $10000\times4 = 4\times10^4$ bps

less susceptible

e) Would 16-QAM be more or less susceptible to noise than this type of modulation?



a: $1\times10^6 \log_2 64$

$= 1\times10^6 \times 6$

$= 6\times10^6$ bps.

12 Consider a channel with a 1 MHz capacity and an SNR of 63.
   a. What is the upper limit to the data rate that the channel can carry?
   b. The result of part (a) is the upper limit. However, as a practical matter, better error performance will be achieved at a lower data rate. Assume we choose a data rate of 2/3 as the maximum theoretical limit. How many signal levels are needed to achieve this data rate?

$4\times10^6 = 2\times1\times10^6 \cdot \log_2 L \quad L=4$

Use Nyquist formula

13 A channel can transmit signals in the frequency range of 2500 kHz to 4000 kHz. Compute its bandwidth. Assuming SNR_{dB} = 40, what will be the theoretical maximum capacity of this channel? 1500kHz

$SNR = 10^4$

$C = 1500\times10^3 \cdot \log_2(1+10^4)$

$C = 1.993 \times 10^7$ bps.

14 A 6-km-long cable has an attenuation of −0.25 dB/km. A signal with a power of 2.84 mW has been received at the end of the cable. Find out the power at which the signal was transmitted from the source. −1.5dB

$-1.5 = 10 \cdot \lg \frac{2.84\times10^{-3}}{P_{in}}$

$10^{-0.15} = \frac{2.84\times10^{-3}}{P_{in}}$

4.01mW

15 How many bits are needed for a QPSK signal? If the baud rate is 500 Mbaud, what will be the bit rate?

16 Given that d = 0.5, what is the required bandwidth for each of the following cases if the data rate is 8000 bps?
   a. ASK
   b. QPSK
   c. 64-QAM

N = 8000 bps

a: r=1 S = 8000 B = 12000

b: r=2 S = 4000 B = 6000

c: r=6 S = $\frac{4000}{3}$ B = 2000